

DETERMINATION OF RARE-EARTH ELEMENTS, YTTRIUM AND SCANDIUM IN ROCKS BY AN ION EXCHANGE-X-RAY FLUORESCENCE TECHNIQUE

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Abstract

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An ion exchange—X-ray fluorescence method for the determination of rare-earth elements (REE), Y and Sc in rocks and minerals is described. Samples are decomposed using a sodium peroxide sinter which dissolves even the most resistant minerals. A two-stage ion-exchange procedure, using hydrochloric and nitric acids, is used to separate other elements from the REE, Y and Sc which are adsorbed onto ion-exchange papers for XRF analysis. Data acquired on the international standard rocks using this technique compare favourably with the accepted values. These results, combined with precision checks on in-house standards, indicate that the precision and accuracy of this technique can be better than or equal to that of other well-established REE analytical techniques.

1. Introduction

Geochemistry of the rare-earth elements (REE) has important applications to the understanding of major petrochemical problems in the earth sciences (for a review see Henderson, 1983). At present a variety of well-established techniques are used for the determination of REE, e.g. neutron activation [both radiochemical (RNAA) and instrumental (INAA)], isotope dilution (ID), thermal mass spectrometry and spark-source

mass spectrometry (SSMS). RNAA and ID require extensive chemical separation procedures (Hooker et al., 1975; Duffield and Gilmore, 1979; Thirlwall, 1982); INAA and SSMS are predominantly instrumental techniques (respec. Potts et al., 1981; Taylor and Gorton, 1977). A new technique, inductively coupled plasma (ICP) emission spectrometry has only recently been applied to the determination of REE (Crock and Litche, 1982).

Two techniques are most commonly used for REE analysis, INAA and isotope dilution thermal mass spectrometry. INAA requires access to an irradiation centre, long intervals over which counting takes place (up to 3 months), and generally yields results for only 9 or 10 of the REE. Isotope dilution thermal mass spectrometry is con-

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sidered to be a reference method in analytical chemistry, capable of yielding results with superior accuracy and precision. A major drawback of this technique for REE analysis is that it requires extensive mass spectrometry (up to 8-hr. machine time per analysis). Both techniques use expensive equipment, small sample sizes and generally restrict REE analyses to specialized studies and/or institutes.

X-ray fluorescence (XRF) techniques for analysis of REE on undiluted rock powders (Norrish and Chappell, 1977) allow the determination of only La, Ce, Pr, Nd, Y and Sc. The levels of accuracy and precision for the REE are generally poor due to low abundances and interferences.

Eby (1972) and Fryer (1977) have described ion exchange—XRF techniques for the analysis of REE. Preconcentration and separation of the REE prior to XRF analysis increases precision, accuracy and eliminates some of the interference problems. Their method has been used successfully in a number of studies (Fryer, 1977; Jenner et al., 1979; Taylor and Fryer, 1980). However, our experience (and that of B.J. Fryer, pers. commun., 1983) suggests that these techniques are highly analyst dependent and can suffer from relatively poor precision (predominantly as a result of poor sample dissolution or later precipitation of insoluble complexes).

We describe here a modified ion exchange—XRF technique, developed at the University of Tasmania, which allows rapid determination of up to 12 REE with a precision and accuracy better than or equal to that of INAA techniques. Major improvements compared to the previous method (Fryer, 1977) include sample decomposition by sodium peroxide sintering, ion-exchange separation of Ba and increased analytical sensitivity for both light REE (LREE) and heavy REE (HREE) by use of better instrumentation. Analytical data for eight international standards (BCR1, GSP1, G2, AGV1, BHVO1,

MAG1, JB1 and JG1) and two internal standards are presented to demonstrate the accuracy and precision of the method.

2. Experimental

2.1. Reagents

Analytical grade Merck® GR sodium peroxide, Univar® hydrochloric (HCl) and nitric acid (HNO₃) were used. Water was single-distilled. The ion-exchange resin used in both sets of columns was Bio-Rad® AG-50W-X8 (100–200 mesh), which is a sulphonated polystyrene cation exchanger. Whatman® SA-2 ion-exchange resin loaded paper was used as the final ion collector and support for the XRF thin-film portion of the procedure. The resin in this ion-exchange paper is Amberlite® IR120, a strong-acid cation exchanger (Campbell et al., 1966). REE and other standards were prepared from Spex® high-purity (> 99.99%) oxides (heated to 900°C before use), metals and carbonates.

2.2. Sample dissolution

Four grams of sodium peroxide were weighed into a Pt crucible along with 1 g of finely ground (< 200 mesh) sample powder. The powders were mixed thoroughly and the crucible was placed in a muffle furnace, maintained at $480 \pm 10^\circ\text{C}$, for 1 hr.

After cooling to room temperature, water was carefully added to the crucible, a little at a time, until the vigorous reaction ceased. The sinter cake was transferred quantitatively to a centrifuge tube where it was washed and centrifuged twice with 50 ml distilled water to remove sodium and silica salts. The Pt crucible was rinsed with a few ml of 1 M HCl, which was transferred to the centrifuge tube. 50- μg -Tm spike were added to the sinter cake and the mixture dissolved in less than 20 ml (total) of 1 M HCl.

2.3. Ion-exchange separation

Two sets of ion-exchange columns made of borosilicate glass (Pyrex®) were used in the separation procedure:

(1) *First stage* — for separation of the major elements from the REE and Ba. Columns — 30-cm length $\times$ 1.1-cm internal diameter, with 18 cm of AG50W-X8® resin (flow rate 2 ml min.⁻¹). Columns were cleaned with 200 ml 6 M HCl and re-equilibrated with 150 ml H₂O.

(2) *Second stage* — for separation of Ba from the REE. Columns — 12-cm length $\times$ 0.7-cm internal diameter, with 7.5 cm of AG50W-X8® resin (flow rate 1 ml min.⁻¹). Columns were cleaned with 35 ml 8 M HNO₃ and re-equilibrated with 30 ml H₂O.

The 1 M HCl sample digest was loaded on the first-stage columns and the major elements eluted with 100 ml 2 M HCl. The eluate could be discarded, or kept for atomic absorption analysis. The REE, Ba, Y and Sc were then stripped off the columns with 200 ml 6 M HCl. This volume was evaporated down to a few ml and quantitatively transferred to a small beaker, after which the solution was evaporated to dryness.

A few ml of 8 M HNO₃ were added to the beaker and the solution evaporated to dryness. The sample was taken up in 5 ml 2 M HNO₃ and loaded on the second-stage columns. An additional 25 ml 2 M HNO₃ (total 30 ml) were used to elute the Ba and any residual Ca, Al and Sr. REE, Y and Sc were collected with 35 ml 8 M HNO₃.

The nitric solution was evaporated to dryness, a few ml of 6 M HCl added and the solution again evaporated to dryness. 15 ml 0.1 M HCl were added to the beaker and heated for a few minutes to dissolve the residue. On cooling to room temperature a disk (31 mm diameter) of ion-exchange paper was added and allowed to equilibrate with the solution for 24 hr. with occasional shaking of the beaker. The solution was evaporated slowly under a heat lamp over a further 24-hr. period and the dried ion-

exchange paper stored in a small paper envelope prior to X-ray analysis.

2.4. Instrumentation

The XRF spectrometer used in this study was a Philips® PW 1410. Operation of the XRF spectrometer had been automated and goniometer angles, counting time, crystal changer, collimator, and kV and mA for 4 samples at a time were controlled by a Radioshack Tandy® TRS-80 microcomputer. Two 5.25-in. (~ 13.3 cm) floppy disks were used for storage of programs and data.

The sample holders for the REE papers had a 28-mm opening and Al masks were used on both sides of the paper to hold it in place. A Nylon® insert was used to back the ion-exchange paper. Papers were counted on each side. Operating conditions for the XRF spectrometer were as follows:

Au target X-ray tube
50 kV, 50 mA
LiF 200 crystal
Fine collimator (150 μ m)
Gas-flow proportional counter, P10 gas + scintillation counter
Pulse-height analyser (lower level 1.2 V, window 1.5 V)
Vacuum
Counting time 40 s per element per side

2.5. Calibration standards and yield correction

Ion-exchange papers for each of the REE, Y, Sc, Ba, Fe, Ni, Cr and Mn were made from solutions prepared from the Spex® HiPure chemicals. For each element the average result from determinations on six separate 200- μ g papers, equilibrated in 15 ml 0.1 M HCl, was used to determine the net count rate (counts s⁻¹ μ g⁻¹) and inter-element correction factors.

After counting the standard paper on both sides, average counts s⁻¹ were calculated and the background, measured on blank papers, was subtracted. The analytical lines,

2θ angles, counts $\text{s}^{-1} \mu\text{g}^{-1}$, detection limits (3σ , 99% confidence) and results from Eby (1972) are listed in Table I.

It is necessary to use many inter-element correction factors; however, the majority of these are small (Table II). The correction factors were obtained from the single-element papers by measuring intensities at all the analytical lines. For example: measurements on the 200- μg standard La paper showed La interfered with background (B1), Pr, Nd, Sm, Eu, Gd and Tb. Correc-

tion factors were worked out as follows, e.g.:

(La correction on Pr) =

$$\frac{(\text{background-corrected counts on Pr line})}{(\text{background-corrected counts on La line})}$$

and applied:

(La correction on Pr) =

$$(\text{La net counts}) \times (\text{correction factor})$$

TABLE I

Analytical lines, count rates and detection limits

Line	2θ (deg)	This work* ¹ (counts $\text{s}^{-1} \mu\text{g}^{-1}$)	l.l.d. (μg)	Eby (1972)* ² (counts $\text{s}^{-1} \mu\text{g}^{-1}$)	l.l.d. (μg)
Sc- $K\alpha_1$	97.65	15.44	0.17	11.56	0.39
Ba- $L\alpha_1$ * ³	87.15	9.98	0.37	—	—
La- $L\alpha_1$	82.90	11.59	0.35	2.91	0.66
B1	81.00	—	—	—	—
Ce- $L\alpha_1$	78.99	14.05	0.33	3.91	0.70
Pr- $L\alpha_1$	75.40	15.87	0.31	4.70	0.49
Nd- $L\alpha_1$	72.11	22.00	0.26	5.40	0.65
Cr- $K\alpha_1$ * ³	69.34	69.90	0.15	—	—
Sm- $L\alpha_1$	66.22	31.57	0.23	7.90	0.43
Eu- $L\alpha_1$	63.57	35.00	0.22	7.50	0.57
Mn- $K\alpha_1$ * ³	62.95	97.84	0.10	—	—
B2	62.20	—	—	—	—
Gd- $L\alpha_1$	61.09	43.60	0.20	8.51	0.47
Tb- $L\alpha_1$	58.77	48.50	0.20	10.38	0.46
Fe- $K\alpha_1$ * ³	57.52	127.83	0.28	—	—
Dy- $L\alpha_1$	56.59	54.10	0.19	10.38	0.46
B3	55.30	—	—	—	—
Ho- $L\alpha_1$	54.54	63.20	0.18	10.67	0.47
Er- $L\alpha_1$	52.61	72.30	0.17	10.70	0.51
Tm- $L\alpha_1$	50.79	74.93	0.17	6.95	0.90
Yb- $L\alpha_1$	49.05	83.90	0.17	6.66	1.13
B4	48.00	—	—	—	—
Lu- $L\alpha_1$	47.41	92.00	0.15	7.00	1.13
Ni- $K\beta_1$ * ³	43.73	35.03	0.58	—	—
Y- $K\alpha_1$	23.78	55.58	0.49	10.16	1.48

B1, B2, B3 and B4 are background positions.

l.l.d. = lower limit of detection, 3σ 99% confidence level calculated using:

$$\text{l.l.d.} = 6m^{-1} (\text{Rb}/T)^{1/2}$$

where m = counts $\text{s}^{-1} \mu\text{g}^{-1}$; Rb = counts s^{-1} on background; and T = 80 s.

*¹ This work using Au X-ray tube, LiF 200 crystal, gas-flow proportional + scintillation counter.

*² Eby (1972) using tungsten X-ray tube, LiF 200 crystal, and gas-flow proportional counter (except for Sc- $K\alpha_1$ where Cr tube and PE crystal used).

*³ Residual Ba, Cr, Mn, Fe, Ni after ion exchange.

Ba, Fe, Ni, Cr and Mn are almost entirely removed during the separation procedure. Nonetheless, lines for these elements are monitored, residual concentrations determined (usually $< 10 \mu\text{g}$) and correction made. High residual concentrations of any of these elements indicates either contamination or a poor chemical separation.

Standard (counts $\text{s}^{-1} \mu\text{g}^{-1}$) and background (counts s^{-1}) count rates were stored in the computer program and checks on these count rates indicated very little change over a period of several months. As a regular background check, 4 positions ($B1 = 81.00^\circ$, $B2 = 62.20^\circ$, $B3 = 55.30^\circ$ and $B4 = 48.00^\circ$) evenly spaced between the REE lines were measured in each sample run. The net count rate at these positions was expected to be equivalent to below the detection limit of the REE. If this was not the case a check on blank paper backgrounds and interference corrections was made at all positions.

Correction for loss of REE during the separation procedure was made by monitoring Tm (Fryer, 1977). Sample solutions were spiked with $50 \mu\text{g}$ Tm, after the sodium peroxide sinter. Count rates on standard $50\text{-}\mu\text{g}$ -Tm papers were compared with those found for Tm in the samples, after background and interference corrections. Expected yields were 90–100% and samples outside this range were discarded. Tm was chosen as an internal standard because of its low natural abundance.

Calibration using the pure-element papers was used to analyse the international standard rocks BCR1, AGV1, G2 and GSP1. Agreement was generally within 10% except for the low-abundance elements Tb, Ho and Lu. Consideration of the uncertainties in the stoichiometry of the REE-oxides and in the preparation of the standards led us to adopt a calibration for the REE based on published abundances of the REE in the four international standards mentioned above. This approach is similar to that followed by Potts et al. (1981). Using the results published in Potts et al. (1981) and Gladney et al. (1983) a new calibration

was established and checked using two internal standards (basalt — TASBAS and granite — TASGRAN) and four international standards (BHVO1, MAG1, JB1 and JG1).

3. Discussion

The procedure presented in this paper was arrived at after a number of possibilities were considered or tried. A discussion of some of the problems and tests made is given prior to presentation of the results.

3.1. Sample dissolution

Acid digestions using HF and HClO_4 are time consuming and potentially dangerous, both to the analyst and the laboratory. Not uncommonly there are insoluble fluorides left, which depending on the sample chemistry may contain REE. Complete dissolution of resistant minerals such as zircon, tourmaline or monazite, which often contain a large fraction of the sample's REE content, is not possible without use of sealed PTFE digestion vessels. This is a time-consuming process, some samples require 3–7 days, and still involves the use of HF and HClO_4 .

A lithium metaborate fusion will give complete dissolution of the sample; however, sample size is limited, HF must be used to remove the silica and boric acid precipitates when using HCl to dissolve the fusion cake (Crock et al., 1984). The procedure using a potassium bifluoride fusion is lengthy and involves the use of HF (Brenner et al., 1984).

A sodium peroxide sinter involves only 1 hr. in the furnace, two water washes and dissolution of the cake easily in 1 M HCl. No HF or HClO_4 is used in the procedure and up to 5 g of sample have been decomposed. The sinter renders even the most resistant minerals soluble (Rafter, 1950; Seelye and Rafter, 1950).

Some care is necessary in using the sodium peroxide method:

- (1) It is necessary to maintain the tem-

perature in the muffle furnace accurately. At temperatures in excess of 500°C the Pt crucible is rapidly corroded (Clarke, 1961). Our experience is that 480°C is sufficient and leads to a minimal loss of Pt. Ni or Fe crucibles are reported to have a more rapid erosion rate (Belcher, 1963).

(2) It is important to ensure the sodium peroxide and the finely ground sample are well mixed. A sample/ Na_2O_2 ratio of 1 : 4 was found to be satisfactory. It is possible to vary this ratio and we have brought 5 g of sample into solution using only 10 g Na_2O_2 .

3.2. Blank levels

Potentially the most serious source of contamination is from the sodium peroxide (Belcher, 1963). To check this, a sample from each new batch of sodium peroxide was analysed for REE by dissolving it in water, neutralising with HCl and processing it through the ion-exchange procedure for XRF analysis. No serious source of impurity was detectable. The other potential source of contaminants are the acids and water. The water was found to be free of all contaminants. The only significant contaminant in the acids is Fe, which is present at the level of 0.05–0.3 $\mu\text{g ml}^{-1}$ depending on the brand used. Fe is introduced during the elution of the REE on the first-stage columns and is not removed on the nitric columns. Originally, vapour-distilled HCl was used to remove this; however, this technique is time consuming and each new batch of acid is now monitored and inter-element effects removed by the data reduction program.

A reagent blank was also run by substituting Spex® HiPure silica for the sample in the full ion-exchange–XRF procedure. The REE, Y and Sc contents were found to be negligible.

3.3. Mass absorption

Mass absorption effects of REE ions on SA-2® ion-exchange papers were calculated

by Eby (1972). For concentrations of REE and Zr of < 500 μg on the paper absorption effects are < 5%. Increasing amounts of these ions on the paper leads to depression of count rates, for example with 1000 μg there is a 10% reduction. Therefore, it is necessary to reduce the sample weight used in those samples with high REE abundances. The use of Tm for yield correction compensates for absorption, to the extent that any depression in count rate of the REE also depresses the Tm count rate. The resultant absorption effects after the yield correction are negligible.

3.4. Ion-exchange procedure

The first-stage ion-exchange procedure is an adaption of previously described techniques (Schnetzler et al., 1967; Eby, 1972; Fryer, 1977; Strelow and Jackson, 1974). The major elements and some trace elements are eluted using 2 M HCl, leaving Sc, Y, REE, Ba, Th, and some Zr and Sr on the columns. The retained elements are eluted using 6 M HCl, in the order Sr, Sc, Y, Ba, REE (Lu through La) and Zr.

To ensure a complete and unfractionated recovery of the REE two factors are crucial: (1) the correct volumes of acid must be used to ensure no REE are lost in the discarded portion and that all the REE are later eluted from the columns; and (2) the columns must not be overloaded. The first-stage columns described here are just large enough to give an adequate separation for a 1-g rock sample. A further note of caution, even samples of this size may overload the columns if there is an abundance of a particular cation, e.g. Al in anorthosites. Major-element compositions and suspected REE patterns are generally known before the REE are analysed. The sample size can then be adjusted so as not to overload the columns. An unknown sample should preferably be analysed for Y by the normal XRF pressed powder pill method (Norrish and Chappell, 1977) from which an approximate idea of REE content is obtained. If there is doubt

over the major-element composition, e.g. excessive Al, then a 0.5-g sample instead of the usual 1 g should be used. It is also possible to use larger columns and/or finer resin to increase the capacity; however, larger volumes of acid are required and a slower separation will result. For example, we have processed up to 5 g of sample using a 18×2.3 cm column loaded with 200–400 mesh AG50W-X8[®]. The discard acid fraction was 500 ml 2 M HCl and the REE were eluted with 600 ml 6 M HCl. The flow rate was 3.0 ml min.⁻¹.

To calibrate the first-stage columns the following procedure was used:

(1) A 1-g sample of our in-house basalt standard (TASBAS) was digested and spiked with 100 μ g Lu. Lutetium is the first of the REE eluted (at virtually the same time as Sc and Y) and Al is the last of the major elements eluted. 10-ml aliquots of 2 M HCl were collected, evaporated to dryness, taken up in 0.1 M HCl and evaporated onto ion-exchange papers for XRF analysis. 100 ml 2 M HCl removed 99.4% of the Al without any elution of Lu.

(2) Elution curves for La, the most strongly retained of the REE, were prepared by loading 100 μ g La on the columns and eluting with 4 M and 6 M HCl. Strelow and Jackson (1974) used 4 M HCl because it was found the distribution coefficient of La did not decrease appreciably in acids above this strength. Our results on the 18×1.1 cm column indicated an extensive tail with La and 200 ml 6 M HCl or 250 ml 4 M HCl were needed for total recovery. Crock et al. (1984) also noted a La tail, eluting the La with 160 ml 8 M HCl in a 20×1 cm column of 100–200 mesh AG-50W-X8[®] resin.

The second-stage columns use HNO₃ acid to achieve a separation of Ba and the REE. Ba interferes with Ce during XRF analysis and can overload the ion-exchange paper. The columns also serve to clean up any residual Ca, Al and Sr in the solution. Elution experiments to determine the amount

of discard were made using a solution of 1000 μ g Ba and 200 μ g Lu. The optimum volume was determined to be 30 ml 2 M HNO₃. Using a solution of 100 μ g La it was determined that 35 ml 8 M HNO₃ will recover all the REE.

To maintain the thin-film characteristics of the ion-exchange paper and reduce interferences it is preferable to remove all the ions other than the REE, Y and Sc. Strelow et al. (1965, 1971), Strelow (1966) and Strelow and Jackson (1974) have shown that HCl provides a better separation for the REE than HNO₃. Crock et al. (1984) preferred to use HNO₃ because it is possible to elute the REE with less volume of acid (the LREE will form nitrate ion association complexes). If HNO₃ is used then a second anion-exchange procedure is necessary to remove Fe from the REE. Crock et al. (1984) report early elution of Al and Zr in their technique. These findings are not in agreement with the determined distribution coefficients and F.W.E. Strelow (pers. commun., 1984) suggests this is due to the formation of very stable Al- and Zr-fluoride complexes after an acid sample digestion.

An alternative which is presently being investigated uses advantages of both HCl and HNO₃. The majors are eluted with 2 M HCl and the REE with 5 M HNO₃ on the same column. Only 130 ml 5 M HNO₃ are required to remove the REE compared with 200 ml 6 M HCl. Higher concentrations of HNO₃ are not advisable because the resin is attacked slightly at 6 M HNO₃ and slowly loses capacity. This attack becomes more pronounced with larger concentrations of HNO₃.

3.5. Instrumentation

Use of either a Au or W X-ray tube is necessary (Norrish and Chappell, 1977). Our results with a Au tube (Table I) show a substantial increase in sensitivity compared to those of Eby (1972), who used a W tube. This is more likely due to modern instrumentation and increased resolution rather than

differences in the X-ray tubes. The optimum tubes for Y and Sc are Mo and Cr respectively, however, good detection limits were obtained with the Au tube.

Combining use of the gas-flow proportional counter and the scintillation counter gives a marked increase in count rates for the HREE. The use of the scintillation counter is essential for determination of Y, but for the HREE there are only relatively small increases in precision and improvements in detection limits due to consequent increases in background counts s^{-1} .

The variation in intensity between the top and bottom of the papers is usually less than 10% and no systematic differences have been noted.

4. Results

4.1. Precision

Triplicate analyses of our in-house standards, TASBAS and TASGRAN, were made to determine the precision or intra-laboratory reproducibility (Table III; Fig. 1).

TABLE III

Precision of TASBAS and TASGRAN analysed in triplicate

Element	TASBAS		TASGRAN	
	(ppm)	σ (%)	(ppm)	σ (%)
La	43.2 $\pm$ 1.25	2.9	38.8 $\pm$ 0.93	2.4
Ce	89.1 $\pm$ 2.58	2.9	86.9 $\pm$ 2.00	2.3
Pr	10.34 $\pm$ 0.28	2.7	9.96 $\pm$ 0.25	2.5
Nd	41.8 $\pm$ 0.79	1.9	37.4 $\pm$ 0.56	1.5
Sm	8.15 $\pm$ 0.10	1.2	7.09 $\pm$ 0.11	1.6
Eu	2.61 $\pm$ 0.11	4.3	0.83 $\pm$ 0.10	12.5
Gd	7.01 $\pm$ 0.10	1.5	6.14 $\pm$ 0.23	3.8
Tb	1.04 $\pm$ 0.06	5.4	1.04 $\pm$ 0.04	3.8
Dy	4.89 $\pm$ 0.10	2.0	5.98 $\pm$ 0.16	2.6
Ho	0.90 $\pm$ 0.15	16.7	1.30 $\pm$ 0.05	3.9
Er	2.05 $\pm$ 0.15	7.2	3.78 $\pm$ 0.06	1.5
Yb	1.26 $\pm$ 0.06	4.6	3.40 $\pm$ 0.06	1.7
Lu	<0.15		0.5 $\pm$ 0.11	22.0
Y	23.5 $\pm$ 0.73	3.1	36.3 $\pm$ 0.69	1.9
Sc	14.1 $\pm$ 0.06	0.4	6.7 $\pm$ 0.08	1.2

σ (%) = standard deviation as per cent.

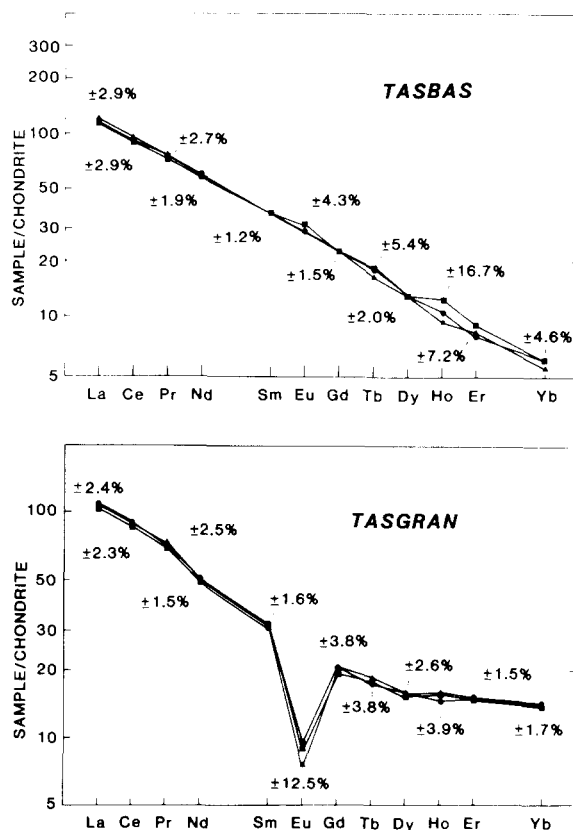


Fig. 1. Triplicate analyses of TASBAS and TASGRAN. Precision in per cent.

Precision for La, Ce, Pr, Nd, Sm, Gd, Dy, Y and Sc is 1–4%. The results for Eu, Er and Yb are more variable because of their relatively low abundances. Abundances of Ho, Tb and Lu are difficult to determine with the XRF method. Ho has a large interference correction from Gd which compounds the problems due to its low abundance. Our experience with Tb is that we generally cannot do better than ± 5 –15%, although we did do so with our in-house standards. Lu is often too close to the detection limit ($0.15 \mu g g^{-1}$) for determination. In general, we do not use the Ho, Tb or Lu results.

According to Henderson and Pankhurst (1983) it is difficult to achieve precision better than 2% using INAA and they suggest the following guidelines for a basalt like BCR1 2–4% for La, Ce, Nd, Sm, Eu

TABLE IV (continued)

	XRF	ID ¹	ID ²	NAA ¹	NAA ²	SSMS	ICP		SURVEY	
							(1)	(2)		
G2:										
La	86.0	—	97.7, 96	89.9	84	—	—		86	± 5
Ce	165.5	—	160, 150	164.0	155	—	—		159	± 11
Pr	15.2	—	—	15.7*	—	—	—		19	± 2
Nd	52.4	—	54.8, 60	53.4	51	—	—		53	± 8
Sm	7.54	—	7.27, 7.3	7.0	7.1	—	—		7.2	± 0.6
Eu	1.63	—	1.34, 1.50	1.38	1.27	—	—		1.41	± 0.12
Gd	4.13	—	3.97, 5.0	4.47*	3.7	—	—		4.1	± 0.8
Tb	0.55	—	—	0.58	0.55	—	—		0.48	± 0.07
Dy	2.15	—	2.11, 2.60	2.81*	—	—	—		2.5	± 0.5
Ho	—	—	—	0.49*	0.46	—	—		0.37	± 0.02
Er	1.00	—	0.83, 1.30	1.13*	—	—	—		1.2	± 0.3
Yb	0.81	—	0.60, 0.38	0.78	0.74	—	—		0.78	± 0.14
Lu	< 0.15	—	—	0.10*	0.094	—	—		0.11	± 0.02
Y	10.8	—	—	—	—	—	—		11.4	± 2.3
Sc	3.5	—	—	—	—	—	—		3.5	± 0.4
GSP1:										
La	162	—	182, 191	179.2	168	—	188		183	± 13
Ce	405	—	431, 419, 394	440.8	429	—	432		406	± 20
Pr	52.8	—	—	50.7*	—	—	56		51	± 8
Nd	185	—	204, 201, 188	202.5	203	—	206		190	± 17
Sm	25.5	—	26.4, 25.8, 27.1	25.1	26.3	—	27.4		26.8	± 2.5
Eu	2.11	—	2.23, 2.21, 2.4	2.30	2.17	—	2.16		2.36	± 0.22
Gd	11.9	—	12.0, 10.2, 15.0	12.9*	11.7	—	12.4		13	± 2
Tb	1.57	—	—	1.63	1.91	—	0.4		1.36	± 0.14
Dy	5.43	—	5.52, 5.45, 5.40	7.9*	—	—	5.8		5.4	± 0.4
Ho	0.94	—	—	1.3*	1.14	—	1.00		1.2	± 0.5
Er	2.72	—	2.14, 2.11, 3.0	2.9*	—	—	1.7		2.5	± 0.4
Yb	1.87	—	1.44, 1.50, 1.80	1.77	1.75	—	1.4		1.7	± 0.4
Lu	< 0.15	—	—	0.21*	0.21	—	0.16		0.22	± 0.05
Y	31.8	—	—	—	—	—	24.0		29	± 6
Sc	6.2	—	—	—	—	—	—		6.1	± 0.5

XRF = this work, ion exchange—X-ray fluorescence spectrometry.

ID¹ = Nakamura (1974), isotope dilution mass spectrometry.

ID² = Hooker et al. (1975), isotope dilution mass spectrometry.

NAA¹ = Potts et al. (1981), instrumental neutron activation analysis.

NAA² = B.J. Chappell (pers. commun., 1984), instrumental neutron activation analysis.

SSMS = Taylor and Gorton (1977), spark-source mass spectrography.

ICP = Crock and Lichte (1982), inductively coupled argon plasma—optical emission spectrometry [(1) = acid digestion; (2) = LiBO₂ fusion].

SURVEY = Gladney et al. (1983). Compilation using ID and NAA, except for Pr and Y where overall mean used.

*Determined by interpolation of chondrite-normalised values.

et al. (1983) calculated “consensus values”, using selected analytical techniques for certain elements (NAA and ID for the REE

and Sc, excluding Pr and Y for which the overall mean was used). ID analysis would seem to be the reference technique (Hender-

TABLE V

Comparison of REE data (ppm) in the U.S.G.S. standard rocks BHVO1 and MAG1

	XRF	ID	NAA ¹	NAA ²	SSMS	ICP	SURVEY	(n)
BHVO1:								
La	16.2	15.55	15.5	15.7	16.7	16.2	17?	(9)
Ce	39.5	38.20	38.9	39.1	41.8	38.2	39±3	(12)
Pr	5.57	—	4.0*	—	5.57	5.66	—	
Nd	26.4	24.75	25.1	24.2	27.8	25.4	24 ⁺² ₋₁	(13)
Sm	6.40	6.16	6.11	5.94	6.34	6.4	6.1±0.8	(12)
Eu	2.16	2.08	2.11	2.12	2.00	2.18	2.0 ^{+0.3} _{-0.2}	(12)
Gd	6.55	6.24	6.1*	—	5.74	7.0	6.0 ^{+0.5} _{-0.4}	(10)
Tb	1.07	—	0.97	0.88	0.86	0.86	1.0?	(8)
Dy	5.78	5.35	5.5*	—	5.02	5.59	5?	(6)
Ho	1.15	—	1.1*	0.95	0.97	1.06	—	
Er	2.66	2.56	2.7*	—	2.40	2.63	—	
Yb	1.98	2.00	2.09	1.97	1.90	2.19	1.9±0.2	(14)
Lu	0.35	0.28	0.30	0.275	0.29	0.29	—	
Y	28.7	—	—	—	30.0	25.7	27?	(9)
Sc	32.8	—	—	32.7	—	—	31 ⁻⁰ ₊₁	
MAG1:								
La	39.5	—	41.8	—	39.3	—	41?	(7)
Ce	82.6	—	87.6	—	91.5	—	86?	(10)
Pr	9.43	—	9.9*	—	10.9	—	—	
Nd	37.3	—	39.1	—	42.6	—	41 ⁺² ₋₆	(11)
Sm	7.61	—	7.22	—	7.96	—	8.1?	(8)
Eu	1.64	—	1.43	—	1.47	—	1.5?	(9)
Gd	6.10	—	6.1*	—	5.55	—	6.6?	(6)
Tb	0.98	—	0.96	—	0.84	—	1.0?	(6)
Dy	5.24	—	5.5*	—	5.02	—	—	
Ho	1.14	—	1.1*	—	1.12	—	—	
Er	2.91	—	3.0*	—	3.16	—	—	
Yb	2.81	—	2.57	—	2.95	—	2.6 ^{+0.3} _{-0.2}	(11)
Lu	0.45	—	0.40	—	0.45	—	—	
Y	31.0	—	—	—	30.8	—	27?	(6)
Sc	16.8	—	—	—	—	—	17±1	(10)

XRF = this work, ion exchange—X-ray fluorescence spectrometry.

ID = M. Rautenschlein and G.A. Jenner (pers. commun., 1984), isotope dilution mass spectrometry.

NAA¹ = Potts et al. (1981), instrumental neutron activation analysis.NAA² = Roden et al. (1984), instrumental neutron activation analysis.

SSMS = BHVO1 (Taylor and Gorton, 1977); MAG1 (McLennan and Taylor, 1980), spark-source mass spectrography.

ICP = Crock and Lichte (1982), inductively coupled argon plasma—optical emission spectrometry.

SURVEY = Abbey (1982), compilation from selected laboratories. (n) = total number of laboratories.

*Determined by interpolation of chondrite-normalised values.

TABLE VI

Comparison of REE data (ppm) in international standards JG1 and JB1

	JG1					JB1					
	XRF	ID	NAA ¹	NAA ³	SURVEY	XRF	ID	NAA ¹	NAA ²	NAA ³	SURVEY
La	21.5	21.48	22.5	21.9	22?	37.2	37.3	36.6	38.6, 37.6	37.7	36?
Ce	47.7	42.85	45.2	45	43?	66.6	67.9	66.9	69.1, 67.1	68	67?
Pr	5.1	—	5.1*	—	—	6.8	—	7.1*	—	—	—
Nd	21.0	20.58	20.2	20	—	27.3	27.0	27.1	27, 28.3	26	21?
Sm	4.45	4.48	4.9?	4.73	4.6?	5.35	5.1	5.0	5.2, 5.3	5.06	4.8?
Eu	0.7	0.708	0.69	0.7	0.7?	1.45	1.54	1.51	1.58, 1.48	1.6	1.5?
Gd	4.25	3.86	4.4*	—	—	4.65	4.80	4.5*	—	—	—
Tb	0.65	—	0.79	0.79	—	0.70	—	0.76	0.84, 0.76	0.65	—
Dy	4.75	3.41	5.2*	—	3?	4.20	4.25	4.5*	—	—	4?
Ho	0.95	—	1.2*	—	—	0.85	—	0.91*	0.81, 0.93	—	—
Er	2.95	1.63	3.3*	—	—	2.35	2.35	2.4*	—	—	2.3?
Yb	3.1	1.48	3.08	3.1	1.5?	1.95	2.15	2.10	1.82, 1.91	2.2	2.1?
Lu	0.3	0.22	0.47	0.46	0.2?	0.25	0.31	0.33	0.28, 0.31	0.31	0.3?
Y	30.9	—	—	—	31?	25.5	—	—	—	—	26?
Sc	6.5	—	—	6.6	6.5?	26.0	—	—	—	28.9	27?

XRF = this work, ion exchange—X-ray fluorescence spectrometry.

ID = N. Nakamura and A. Masuda (pers. commun., 1971), T. Tanaka and A. Masuda (pers. commun., 1971) in Ando et al. (1974), isotope dilution mass spectrometry.

NAA¹ = Potts et al. (1981), instrumental neutron activation analysis.NAA² = Borley and Rogers (1979), instrumental neutron activation analysis.NAA³ = Barnes and Gorton (1984), instrumental neutron activation analysis.

SURVEY = Abbey (1980) compilation from selected laboratories.

*Determined by interpolation of chondrite-normalised values.

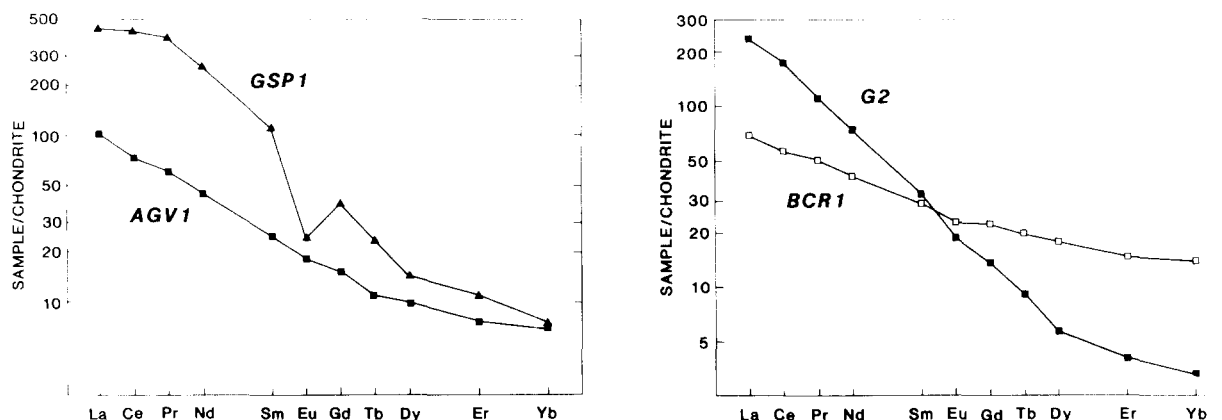


Fig. 2. Chondrite-normalised REE concentrations in AGV1, GSP1, BCR1 and G2 by ion exchange—XRF analysis.

son and Pankhurst, 1983). However, there are two problems: (1) there are not a large number of ID values reported for many of the standards (none in some instances);

and (2) the variation in, or variance from other techniques, of ID results for the granite standards (G2, GSP1 and JG1) is very large (Tables IV and VI). This could be a result

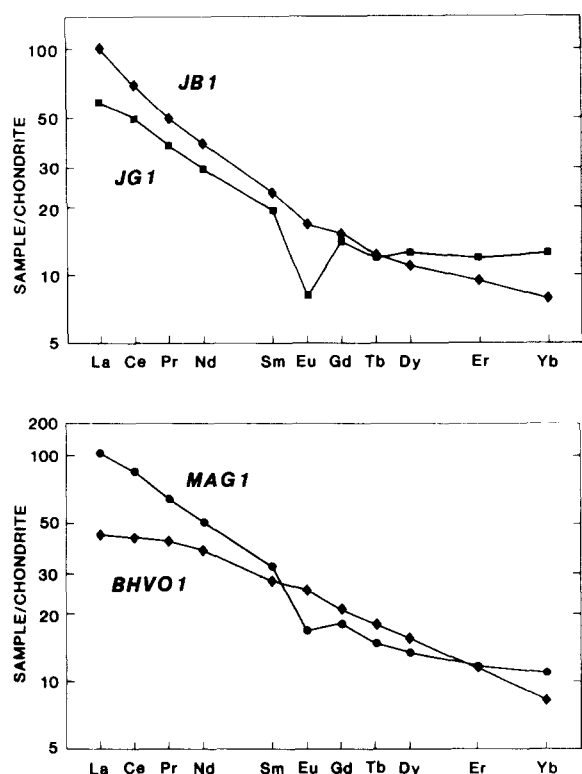


Fig. 3. Chondrite-normalised REE concentrations in JB1, JG1, BHVO1 and MAG1 by ion exchange-XRF analysis.

of sample heterogeneity (Arth and Hanson, 1975), but in large part is probably due to incomplete digestion of the accessory phases in the granites (e.g. Hanson, 1976).

Except for Ho and Lu the XRF values reported for BCR1 and AGV1 in Table IV compare very favourably with those from both ID and INAA. There is < 2% difference between XRF determinations in BCR1 and ID results (Nakamura, 1974), except for La and Dy which differ by 4.3% and 5.5%, respectively. The accuracy for AGV1 is better than 5%. The XRF method appears to give good Pr results but because Pr cannot be determined by ID and NAA there are few published results for comparison. The data for G2 and GSP1 are somewhat more variable, but still compare well with other data reported in Table IV. Far fewer analyses

have been published on G2 (approximately one-third the number compared with BCR1) and even fewer on GSP1 (Potts et al., 1981).

The XRF results for BHVO1, MAG1 and JB1 agree well with the few analyses available by other methods. The granodiorite, JG1 has low ID results (Ando et al., 1974) for the HREE compared with XRF and NAA, which are in good agreement. Although JG1 is a coarse-grained standard, inadequate dissolution of the sample rather than heterogeneity may be the cause of the discrepancy. The ID results were obtained using an acid digestion procedure whereas neutron activation analyses involved no sample decomposition and the XRF method used the powerful sodium peroxide sinter for dissolution.

Table VII and Fig. 4 show a variety of samples with high- and low-REE patterns to illustrate the versatility of the method. The garnet-rich granite is enriched in HREE and is a useful test of inter-element correction factors. Good REE patterns can be obtained down to the 1X chondrite level even though some elements, e.g. Pr, Eu, Tb, are below the detection limits. Care must be taken over background measure-

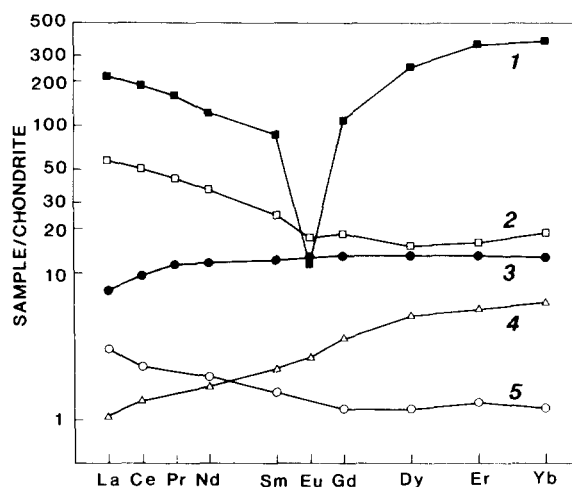


Fig. 4. Chondrite-normalised REE concentrations in: 1 = garnet-rich granite; 2 = dacite; 3 = ocean ridge basalt; 4 = olivine-rich basalt; 5 = dolerite, by ion exchange-XRF analysis.

TABLE VII

REE data (ppm) in a variety of samples analysed by ion exchange—XRF spectrometry

	Garnet-rich granite	Dacite	Ocean ridge basalt	Olivine-rich basalt	Dolerite
La	79.7	21.3	2.84	0.39	1.13
Ce	180.1	49.0	9.23	1.28	2.21
Pr	21.9	6.05	1.56	< 0.31	< 0.31
Nd	86.5	26.2	8.40	1.19	1.40
Sm	20.0	5.76	2.84	0.51	0.36
Eu	1.0	1.51	1.11	0.23	< 0.22
Gd	33.1	5.70	4.04	1.10	0.36
Dy	95.0	5.76	5.03	1.95	0.45
Er	89.0	3.98	3.28	1.41	0.33
Yb	93.5	4.66	3.20	1.56	0.30

ments as well as removal of contamination in the chemical procedure and duplicate analyses must be performed.

5. Conclusions

The ion exchange—XRF method reported in this paper is a valuable geochemical technique which is easily applicable to a wide range of rock types with REE abundances greater than $1\times$ chondrite.

Advantages of the XRF technique are that: (1) it gives values for a more complete spectrum of the REE than does the usual INAA analysis; (2) the sodium peroxide sinter is simple, quick and extremely effective in decomposing resistant minerals. Though not essential, a large sample size (1–5 g) can be used which avoids heterogeneity problems. The REE can be concentrated from 1 g or less of sample, using the ion-exchange procedure which also separates potentially interfering elements; (3) since it requires less specialized, and usually less expensive, equipment it should be available to more geology departments; (4) the total time required for analysis is less than that required for either NAA, SSMS or ID mass spectrometry.

The ion exchange—XRF method has been shown to produce REE data with precision

and accuracy equivalent to that of some of the best INAA data. As would be expected, the XRF cannot produce accuracy and precision found in ID analyses of the basalt standards. However, the XRF method (and INAA) produce better data for the granite standards.

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